Trigonometry & Additional Topics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometry**

In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Adjacent $= \sqrt{20^2 - 12^2} = \sqrt{400 - 144} = \sqrt{256} = 16$. $\sin\theta = \frac{12}{20} = \frac{3}{5}$. $\cos\theta = \frac{16}{20} = \frac{4}{5}$. $\tan\theta = \frac{12}{16} = \frac{3}{4}$.

2 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

Opposite $= \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\sin\theta = \frac{24}{25}$. $\tan\theta = \frac{24}{7}$.

3 **[Foundational]** • **Target: Right-triangle trigonometry**

A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

Height $= 15\sin(50^\circ) \approx 15(0.766) \approx 11.5$ feet.

4 **[Foundational]** • **Target: The Pythagorean identity**

Given $\sin\theta = \frac{5}{13}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\cos\theta = \frac{12}{13}$.

5 **[Foundational]** • **Target: The Pythagorean identity**

Given $\cos\theta = \frac{8}{17}$ with θ acute, find $\sin\theta$.

$\sin^2\theta = 1 - \frac{64}{289} = \frac{225}{289}$. Since θ is acute, $\sin\theta = \frac{15}{17}$.

6 **[Foundational]** • **Target: Exact trigonometric values**

State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.

$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

7 **[Foundational]** • **Target: Exact trigonometric values**

State the exact value of $\tan(60^\circ)$ and $\sin(90^\circ)$.

$\tan(60^\circ) = \sqrt{3}$. $\sin(90^\circ) = 1$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving for missing sides and angles

A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32°, as shown below. Find the height of the tower, to the nearest foot.

$h = 150\tan(32^\circ) \approx 150(0.6249) \approx 94$ feet.

9 [Intermediate] • Target: Solving for missing sides and angles

From the top of a 60-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

The angle of depression equals the angle of elevation from the boat, 25°, so $\tan(25^\circ) = \frac{60}{d}$, giving $d = \frac{60}{\tan(25^\circ)} \approx \frac{60}{0.4663} \approx 129$ feet.

10 [Intermediate] • Target: Complex numbers: arithmetic

Add $(4 + 3i) + (2 - 7i)$.

$6 - 4i$.

11 [Intermediate] • Target: Complex numbers: arithmetic

Multiply $(3 + 2i)(1 - 4i)$.

$3 - 12i + 2i - 8i^2 = 3 - 10i - 8(-1) = 11 - 10i$.

12 [Intermediate] • Target: Complex numbers: arithmetic

Simplify i^7, using $i^2 = -1$.

$i^7 = i^4 \cdot i^3 = 1 \cdot i^3 = i^2 \cdot i = -1 \cdot i = -i$.

13 [Intermediate] • Target: Complex numbers: modulus and conjugates

Find the modulus $|6 - 8i|$.

$\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

14 [Intermediate] • Target: Complex numbers: modulus and conjugates

Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

Conjugate: $-3 - 5i$. Product $= (-3 + 5i)(-3 - 5i) = 9 - 25i^2 = 9 + 25 = 34$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 8.01 Exponentials & Logarithms

Vector $v = (3, 4)$. Find its magnitude.

$|v| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

16 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 1.01 Heart of Algebra

Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.

$u + v = (2 + (-6), -5 + 3) = (-4, -2)$.

17 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 2.01 Quadratics & Polynomials

Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

$2w = (16, -12)$. $|2w| = \sqrt{16^2 + 12^2} = \sqrt{256 + 144} = \sqrt{400} = 20$.

18 [Challenging] • Target: Radian measure • Cross-Lesson: 3.01 Functions & Graphs

Convert 150° to radians, in terms of π.

$150 \times \frac{\pi}{180} = \frac{5\pi}{6}$.

19 [Challenging] • Target: Radian measure • Cross-Lesson: 4.01 Ratios, Rates & Data

Convert $\frac{7\pi}{4}$ radians to degrees.

$\frac{7\pi}{4} \times \frac{180}{\pi} = 7 \times 45^\circ = 315^\circ$.

20 [Challenging] • Target: A well-designed word problem: measuring the width of a river • Cross-Lesson: 5.01 Statistics & Probability

This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C. (a) At point C, the angle between the bank and the line of sight to the tree is 35°. Use the tangent ratio to find AB, the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24°. Use this second measurement to find a second estimate of AB, to the nearest meter, and state which measurement gives the larger estimate.

(a) $\tan(35^\circ) = \frac{AB}{80}$, so $AB = 80\tan(35^\circ) \approx 80(0.7002) \approx 56$ meters. (b) $CB = \frac{80}{\cos(35^\circ)} \approx \frac{80}{0.8192} \approx 98$ meters. (c) $AB = 130\tan(24^\circ) \approx 130(0.4452) \approx 58$ meters. The second measurement (58 m) is slightly larger than the first (56 m); the two estimates are reasonably consistent.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, the hypotenuse is 87 and the side opposite angle θ is 60. Find $\sin\theta$.

a. $\frac{21}{29}$

b. $\frac{20}{29}$ ✓

c. $\frac{20}{21}$

d. $\frac{62}{87}$

$\sin\theta = \frac{60}{87} = \frac{20}{29}$.

22 [Foundational] • Target: The Pythagorean identity

In a right triangle, the hypotenuse is 50 and the side opposite angle θ is 14. Find $\sin\theta$.

a. $\frac{8}{25}$

b. $\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $\frac{7}{24}$

$\sin\theta = \frac{14}{50} = \frac{7}{25}$.

23 [Foundational] • Target: Exact trigonometric values

In a right triangle, the hypotenuse is 25 and the side adjacent to angle θ is 24. Find $\cos\theta$.

a. $\frac{26}{25}$

b. $\frac{24}{25}$ ✓

c. $\frac{7}{24}$

d. $\frac{7}{25}$

$\cos\theta = \frac{24}{25}$.

24 [Foundational] • Target: Solving for missing sides and angles

In a right triangle, the side opposite angle θ is 40 and the side adjacent is 42. Find $\tan\theta$.

a. 1

b. $\frac{21}{29}$

c. $\frac{20}{29}$

d. $\frac{20}{21}$ ✓

$\tan\theta = \frac{40}{42} = \frac{20}{21}$.

25 [Foundational] • Target: Complex numbers: arithmetic

In a right triangle, the hypotenuse is 15 and the side opposite angle θ is 9. Find $\sin\theta$.

a. $\frac{3}{4}$

b. $\frac{11}{15}$

c. $\frac{4}{5}$

d. $\frac{3}{5}$ ✓

$\sin\theta = \frac{9}{15} = \frac{3}{5}$.

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26 [Foundational] • Target: Complex numbers: modulus and conjugates

In a right triangle, the hypotenuse is 51 and the side opposite angle θ is 24. Find $\sin\theta$.

a. $\frac{15}{17}$

b. $\frac{26}{51}$

c. $\frac{8}{15}$

d. $\frac{8}{17}$ ✓

$\sin\theta = \frac{24}{51} = \frac{8}{17}$.

27 [Foundational] • Target: Vectors: magnitude and operations

In a right triangle, the hypotenuse is 10 and the side adjacent to angle θ is 8. Find $\cos\theta$.

a. 1

b. $\frac{4}{5}$ ✓

c. $\frac{3}{4}$

d. $\frac{3}{5}$

$\cos\theta = \frac{8}{10} = \frac{4}{5}$.

28 [Foundational] • Target: Radian measure

Given $\sin\theta = \frac{11}{61}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{62}{61}$

b. $\frac{60}{61}$ ✓

c. 1

d. $\frac{11}{61}$

$\cos^2\theta = 1 - \frac{121}{3721} = \frac{3600}{3721}$. Since θ is acute, $\cos\theta = \frac{60}{61}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: measuring the width of a river

Given $\sin\theta = \frac{9}{41}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{40}{41}$ ✓

b. $\frac{9}{41}$

c. 1

d. $\frac{42}{41}$

$\cos^2\theta = 1 - \frac{81}{1681} = \frac{1600}{1681}$. Since θ is acute, $\cos\theta = \frac{40}{41}$.

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30 [Intermediate] • Target: Trigonometric Ratios & Cofunctions

Given $\cos\theta = \frac{60}{61}$ with θ acute, find $\sin\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{11}{60}$

b. $\frac{60}{61}$

c. $\frac{12}{61}$

d. $\frac{11}{61}$ ✓

$\sin^2\theta = 1 - \frac{3600}{3721} = \frac{121}{3721}$. Since θ is acute, $\sin\theta = \frac{11}{61}$.

31 [Intermediate] • Target: Unit Circle & Radians

Given $\cos\theta = \frac{35}{37}$ with θ acute, find $\sin\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{12}{37}$ ✓

b. $\frac{35}{37}$

c. $\frac{13}{37}$

d. $\frac{1}{3}$

$\sin^2\theta = 1 - \frac{1225}{1369} = \frac{144}{1369}$. Since θ is acute, $\sin\theta = \frac{12}{37}$.

32 [Intermediate] • Target: Complex Numbers & Arithmetic

Given $\sin\theta = \frac{20}{29}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{22}{29}$

b. $\frac{20}{29}$

c. $\frac{21}{29}$ ✓

d. $\frac{3}{4}$

$\cos^2\theta = 1 - \frac{400}{841} = \frac{441}{841}$. Since θ is acute, $\cos\theta = \frac{21}{29}$.

33 [Intermediate] • Target: Coordinate Geometry of Circles

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{4}{5}$ ✓

b. $\frac{3}{5}$

c. $\frac{6}{5}$

d. 1

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos\theta = \frac{4}{5}$.

34 [Intermediate] • Target: Right-triangle trigonometry

Multiply $(8 - i)(1 + 5i)$.

a. $16 + 39i$

b. $13 + 39i$ ✓

c. $3 + 39i$

d. $13 + 41i$

$8 + 40i - i - 5i^2 = 8 + 39i - 5(-1) = 13 + 39i$.

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35 [Intermediate] • Target: The Pythagorean identity

Multiply $(-1-9i)(5+2i)$.

a. $13-47i$ ✓

b. $-23-47i$

c. $13+43i$

d. $13-45i$

$-5-2i-45i-18i^2=-5-47i-18(-1)=13-47i$.

36 [Intermediate] • Target: Exact trigonometric values

Multiply $(7+3i)(-1-8i)$.

a. $-31-59i$

b. $17-59i$ ✓

c. $17-53i$

d. $17-57i$

$-7-56i-3i-24i^2=-7-59i-24(-1)=17-59i$.

37 [Intermediate] • Target: Solving for missing sides and angles

Given vectors $u=(9,\ -8)$ and $v=(5,\ -3)$, find $u+v$.

a. $(-11, 14)$

b. $(14,\ -11)$ ✓

c. $(14,\ -13)$

d. $(16,\ -11)$

$u+v=(9+5, (-8)+(-3))=(14,\ -11)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Complex numbers: arithmetic • Cross-Lesson: 3.01 Functions & Graphs

Given vectors $u=(-4,\ -2)$ and $v=(0, 5)$, find $u+v$.

a. $(-4, 1)$

b. $(3,\ -4)$

c. $(-2, 3)$

d. $(-4, 3)$ ✓

$u+v=((-4)+0, (-2)+5)=(-4, 3)$.

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39 [Challenging] • Target: Complex numbers: modulus and conjugates • Cross-Lesson: 4.01 Ratios, Rates & Data

Convert 120° to radians, in terms of π.

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{2\pi}{3}$ ✓

d. π

$120 \times \frac{\pi}{180} = \frac{2\pi}{3}$.

40 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 5.01 Statistics & Probability

Convert 345° to radians, in terms of π.

a. $\frac{23\pi}{13}$

b. $\frac{23\pi}{6}$

c. $\frac{23\pi}{12}$ ✓

d. 2π

$345 \times \frac{\pi}{180} = \frac{23\pi}{12}$.

41 [Challenging] • Target: Trigonometric Ratios & Cofunctions • Cross-Lesson: 6.01 Geometry

If $\sin(x^\circ) = \cos(42^\circ)$, where $0 < x < 90$, what is the value of x?

a. 42

b. 48 ✓

c. 52

d. 138

By the cofunction identity, $\sin(x^\circ) = \cos(90^\circ - x^\circ)$. Thus $90 - x = 42 \Rightarrow x = 48$.

42 [Challenging] • Target: Unit Circle & Radians • Cross-Lesson: 8.01 Exponentials & Logarithms

Convert 210° to radians.

a. $\frac{5\pi}{6}$

b. $\frac{7\pi}{6}$ ✓

c. $\frac{4\pi}{3}$

d. $\frac{11\pi}{6}$

Multiply by $\frac{\pi}{180^\circ}$: $210 \times \frac{\pi}{180} = \frac{7\pi}{6}$.

43 [Challenging] • Target: Unit Circle & Radians • Cross-Lesson: 1.01 Heart of Algebra

If θ is in Quadrant IV and $\cos\theta = \frac{3}{5}$, what is $\sin\theta$?

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

In Quadrant IV, sine is negative: $\sin\theta = -\sqrt{1 - \cos^2\theta} = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5}$.

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44 [Challenging] • Target: Complex Numbers & Arithmetic • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the simplified form of $\frac{10}{3-i}$?

a. $3+i$ ✓

b. $3-i$

c. $\frac{3+i}{2}$

d. $3+2i$

Multiply by conjugate $(3+i)$: $\frac{10(3+i)}{(3-i)(3+i)} = \frac{10(3+i)}{9-i^2} = \frac{10(3+i)}{10} = 3+i$.

45 [Challenging] • Target: Coordinate Geometry of Circles • Cross-Lesson: 3.01 Functions & Graphs

What is the radius of the circle defined by $(x+4)^2 + (y-7)^2 = 81$?

a. 7

b. 9 ✓

c. 18

d. 81

In $(x-h)^2 + (y-k)^2 = r^2$, $r = \sqrt{81} = 9$.

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